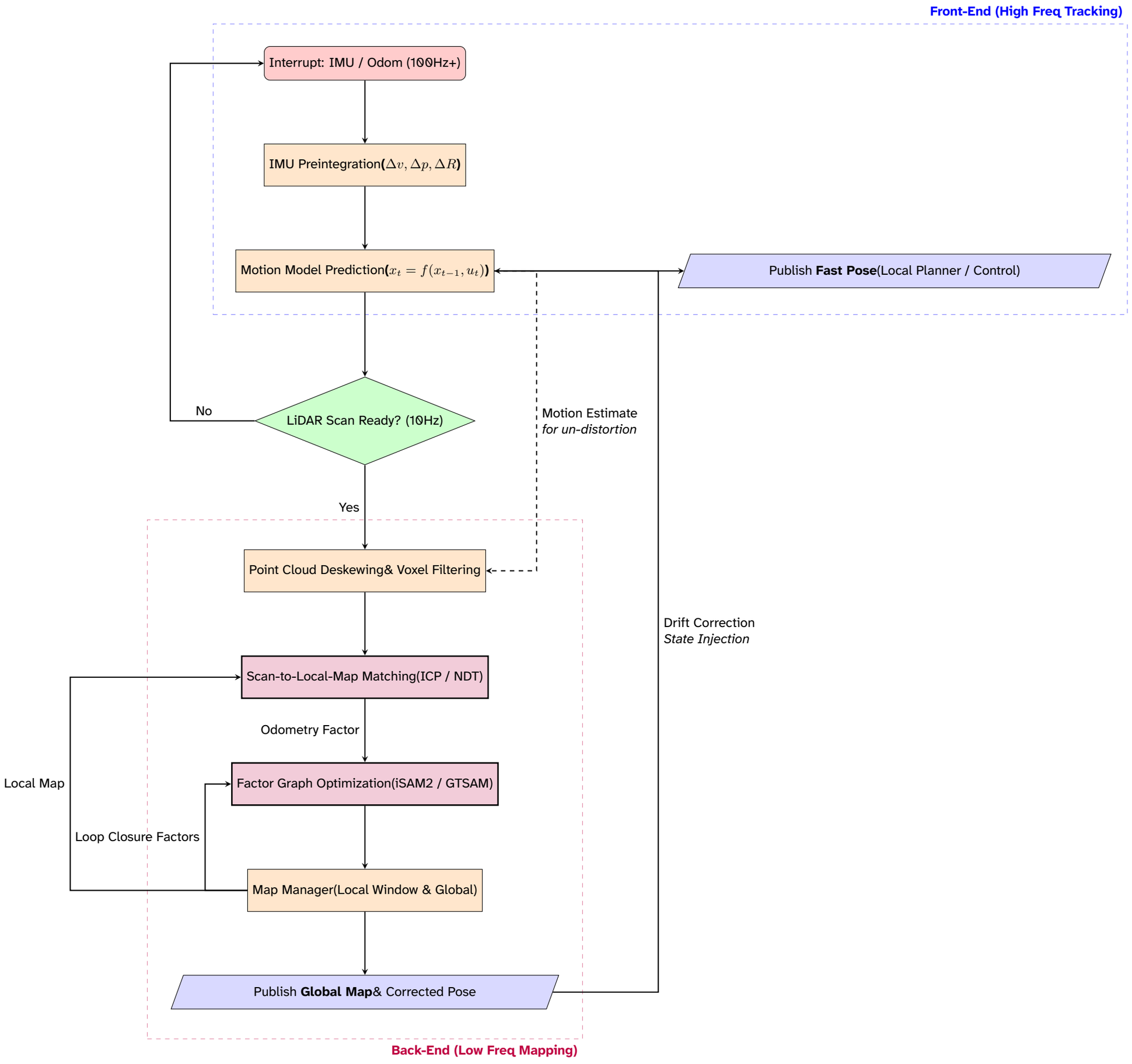
Front-End (High Freq Tracking)
Interrupt: IMU / Odom (100Hz+)
IMU Preintegration($\Delta v, \Delta p, \Delta R$)
Motion Model Prediction($x_t = f(x_{t-1}, u_t)$)
Publish **Fast Pose**(Local Planner / Control)
No
LiDAR Scan Ready? (10Hz)
Motion Estimate
for un-distortion
Yes
Point Cloud Deskewing& Voxel Filtering
Drift Correction
State Injection
Scan-to-Local-Map Matching(ICP / NDT)
Odometry Factor
Local Map
Factor Graph Optimization(iSAM2 / GTSAM)
Loop Closure Factors
Map Manager(Local Window & Global)
Publish **Global Map**& Corrected Pose
Back-End (Low Freq Mapping)